CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Strategic Insights for Stakeholders and Business Leaders

PROJECT OVERVIEW & OBJECTIVES

Primary Objective

To uncover deep insights into customer spending patterns, preferences, and segments to drive strategic growth and enhance operational efficiency.

Scope of Analysis

Comprehensive analysis of **3,900 transactional records** across diverse product categories, encompassing customer demographics, purchase history, and behavioral metrics.

Key Focus Areas

Revenue Generation

Analysis of demographic-driven spending habits and high-value customer segments.

Subscription Models

Evaluating the effectiveness of current subscription and loyalty strategies.

Marketing Impact

Assessing the influence of discounts and shipping preferences on transaction value.

DATA FOUNDATION & METHODOLOGY

INGESTION

- Import 3,900 rows via Pandas
- Structural integrity check
- Summary statistics audit

PROCESSING

- Missing value imputation
- Feature engineering
- Column standardization

ANALYSIS

- PostgreSQL integration
- Structured SQL queries
- Business logic validation

VISUALIZATION

- Power BI dashboards
- Interactive reporting
- Stakeholder insights

3,900	18	Python	SQL
TOTAL RECORDS	KEY FEATURES	TECH STACK	DATABASE

REVENUE INSIGHTS: DEMOGRAPHICS

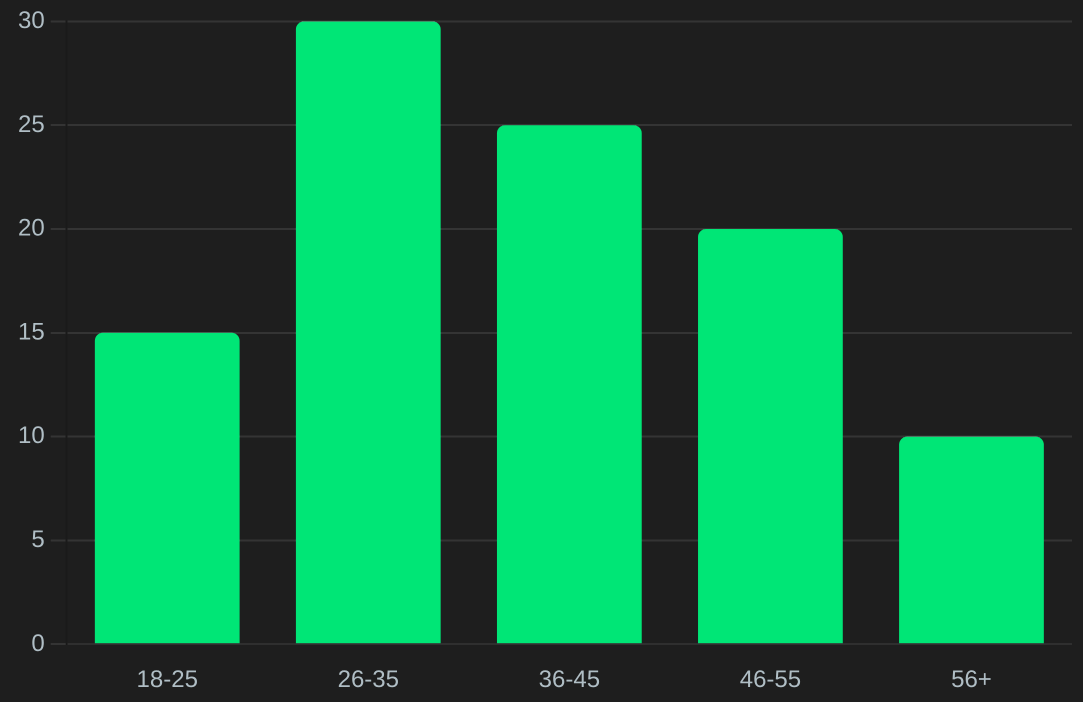
Gender Contribution

Male Customers	**High Volume**
Female Customers	**Premium Spend**

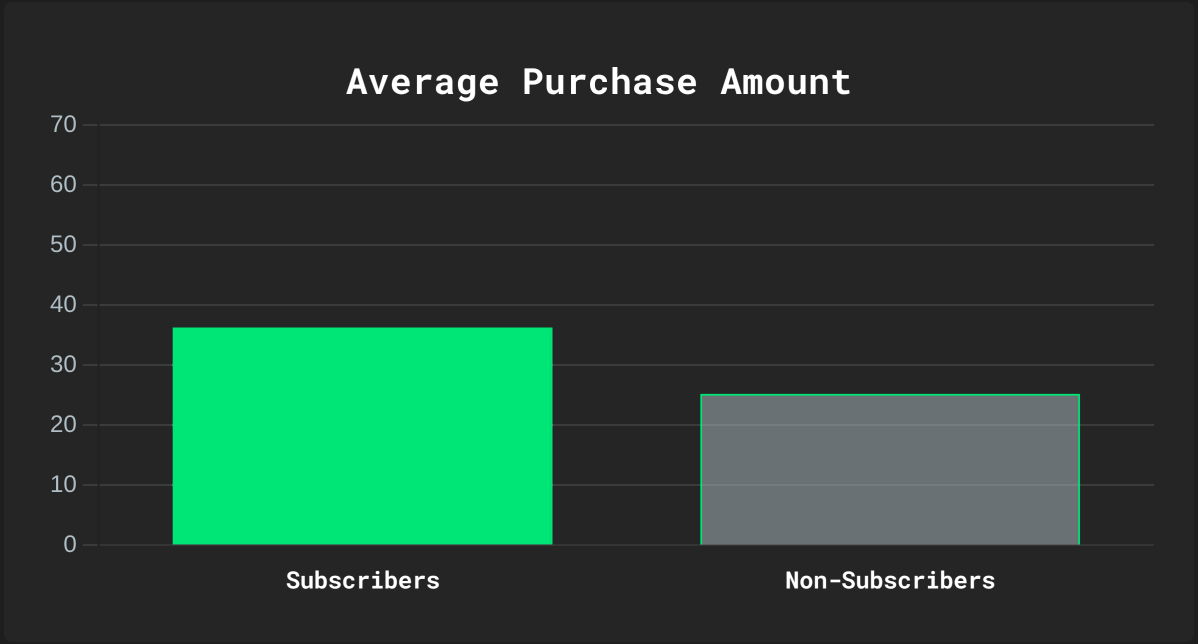
*Comparative analysis of total revenue generated across 3,900 transactions.

Strategic Takeaway: Marketing efforts should be tailored to the specific needs and spending habits of the most profitable demographic groups to maximize ROI.

Revenue by Age Group



SUBSCRIPTION STRATEGY: DRIVING LONG-TERM VALUE



Comparative analysis of average transaction value by subscription status.

Revenue Stability

Subscribers represent a high-value, predictable revenue stream. Data indicates that subscription status is a primary driver of consistent purchase frequency.

High Correlation

LOYALTY & SUBSCRIPTION

Customers with **>5 previous purchases** show a significantly higher propensity to adopt subscription models.

Strategic Goal: Target high-frequency "Returning" customers for subscription conversion to maximize Lifetime Value (LTV).

PRODUCT EXCELLENCE & SATISFACTION

Top 5 Rated Products

PRODUCT NAME	AVG. RATING
Premium Leather Boots	★★★★★ 4.9
Silk Evening Scarf	★★★★½ 4.8
Designer Denim Jacket	★★★★½ 4.7
Wool Blend Sweater	★★★★★ 4.7
Active Performance Tee	★★★★½ 4.6

Quality Benchmark: High review ratings across core categories indicate strong product-market fit and customer satisfaction.

Category Leaders

Apparel

1. Denim Jacket	High Frequency
2. Cotton T-Shirt	Steady Volume
3. Wool Sweater	Seasonal Peak

Footwear

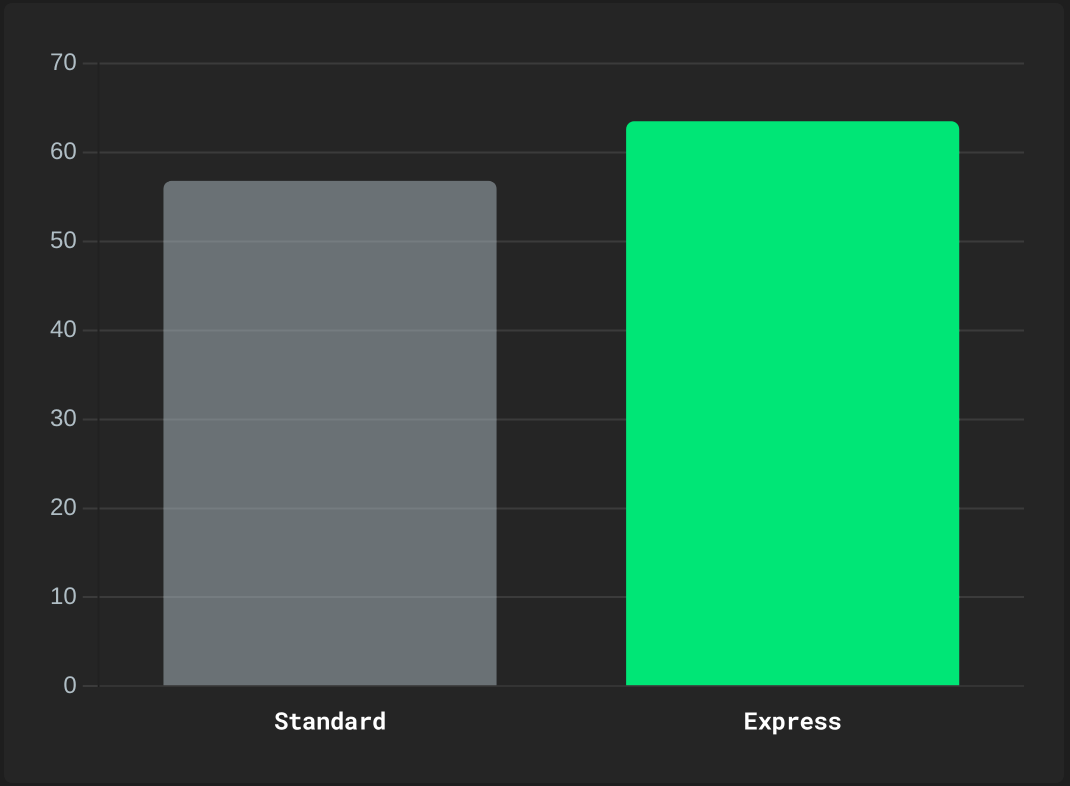
1. Leather Boots	Top Performer
2. Running Shoes	Growth Segment
3. Casual Sneakers	Mass Appeal

Accessories

1. Silk Scarf	Premium Choice
2. Leather Belt	Essential Item
3. Canvas Tote	Eco-Friendly

DISCOUNTS & SHIPPING IMPACT

Shipping Value Comparison



**Average purchase amount ($) by shipping method.*

Discount Efficiency

% High-Spending Discount Users

Analysis identifies a significant segment of customers who utilize discounts but maintain transaction values **above the organizational average**.

Strategic Insight:

Express shipping users correlate with 12% higher transaction values, suggesting an opportunity for premium service positioning and upsell triggers.

CUSTOMER SEGMENTATION: NEW VS. LOYAL

NEW

Recent first-time purchasers who have just entered the brand ecosystem. Focus is on initial satisfaction and welcome engagement.

1 Purchase

RETURNING

Customers with a moderate purchase history. This segment represents the highest potential for migration to long-term loyalty.

2-4 Purchases

LOYAL

High-frequency, long-term brand advocates. These customers provide the most stable revenue and highest lifetime value.

5+ Purchases

Strategic Retention Goal: Develop targeted engagement programs to migrate "Returning" customers into the "Loyal" segment through personalized rewards and subscription incentives.

STRATEGIC RECOMMENDATIONS

BOOST SUBSCRIPTIONS

Launch targeted campaigns promoting **exclusive benefits** for the subscription model. Focus on converting high-frequency buyers who show high brand affinity.

LOYALTY PROGRAMS

Implement tiered rewards for repeat buyers to increase the **"Loyal" customer base**. Gamify the experience to encourage progression from "Returning" to "Loyal".

OPTIMIZE DISCOUNTS

Refine discount policies to **balance volume growth** with margin preservation. Use data to identify products where discounts drive the highest incremental value.

TARGETED MARKETING

Prioritize **high-revenue age groups** and express-shipping users in upcoming campaigns. Personalize messaging based on historical category preferences.

CONCLUSION & NEXT STEPS

STRATEGIC SUMMARY

The analysis of 3,900 transactions provides a clear, data-driven roadmap for enhancing customer lifetime value. By focusing on **segmentation, subscription growth, and product excellence**, the organization is positioned for sustainable revenue expansion.

PHASE 1: INTEGRATION

Deploy real-time Power BI dashboards for ongoing performance monitoring and KPI tracking.

PHASE 2: PILOT PROGRAM

Launch the updated loyalty and subscription pilot in high-performing demographic regions.

PHASE 3: REVIEW

Conduct a comprehensive Q3 results review to measure impact and refine strategic initiatives.